```
clear; clc; close all;

% Design Parameters
Fs = 10000;   % Sampling frequency in Hz
Ap = 0.5;     % Passband ripple in dB
As = 30;      % Stopband attenuation in dB

% ========================================
% 1. CHEBYSHEV TYPE-I LOW-PASS FILTER DESIGN
% ========================================
fp_lp = 1500;  % Passband frequency in Hz
fs_lp = 2000;  % Stopband frequency in Hz

wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);

[N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');

% ========================================
% 2. CHEBYSHEV TYPE-I HIGH-PASS FILTER DESIGN
% ========================================
fp_hp = 2500;  % Passband frequency in Hz
fs_hp = 1800;  % Stopband frequency in Hz

wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);

[N_hp, Wn_hp] = cheb1ord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = cheby1(N_hp, Ap, Wn_hp, 'high');

% ========================================
% DISPLAY NUMERICAL OUTPUTS
% ========================================
fprintf('=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===\n\n');
```

```
=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===
```

```
fprintf('--- CHEBYSHEV LOW-PASS FILTER ---\n');
```

```
--- CHEBYSHEV LOW-PASS FILTER ---
```

```
fprintf('Filter Order (N_lp): %d\n', N_lp);
```

```
Filter Order (N_lp): 6
```

```
fprintf('Cutoff Frequency: %.4f Hz\n', Wn_lp * (Fs/2));
```

```
Cutoff Frequency: 1500.0000 Hz
```

```
fprintf('Numerator Coefficients (b_lp):\n');
```

```
Numerator Coefficients (b_lp):
```

```
disp(b_lp);
```

```
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006
```

```
fprintf('Denominator Coefficients (a_lp):\n');
```

```
Denominator Coefficients (a_lp):
```

```
disp(a_lp);
```

```
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368
```

```
fprintf('\n--- CHEBYSHEV HIGH-PASS FILTER ---\n');
```

```
--- CHEBYSHEV HIGH-PASS FILTER ---
```

```
fprintf('Filter Order (N_hp): %d\n', N_hp);
```

```
Filter Order (N_hp): 6
```

```
fprintf('Cutoff Frequency: %.4f Hz\n', Wn_hp * (Fs/2));
```

```
Cutoff Frequency: 2500.0000 Hz
```

```
fprintf('Numerator Coefficients (b_hp):\n');
```

```
Numerator Coefficients (b_hp):
```

```
disp(b_hp);
```

```
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117
```

```
fprintf('Denominator Coefficients (a_hp):\n');
```

```
Denominator Coefficients (a_hp):
```

```
disp(a_hp);
```

```
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650
```

```
% ==========================================
% FREQUENCY RESPONSE CALCULATION
% ==========================================
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);

% Avoid log10(0)
mag_lp = 20*log10(max(abs(H_lp), 1e-10));
mag_hp = 20*log10(max(abs(H_hp), 1e-10));
```

```
% ==========================================
% PLOT FREQUENCY RESPONSES
% ==========================================
figure('Name', 'Chebyshev Type-I LPF and HPF Responses');

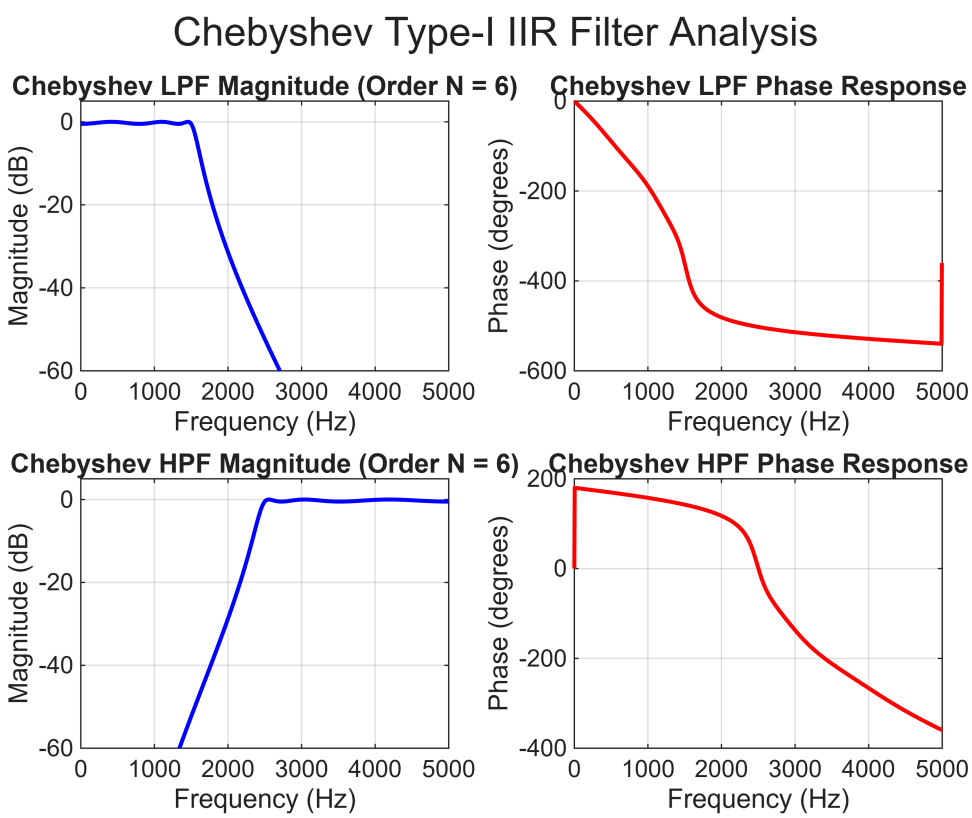
% Low-Pass Magnitude Response
subplot(2,2,1);
plot(f, mag_lp, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev LPF Magnitude (Order N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);
xlim([0 Fs/2]);

% Low-Pass Phase Response
subplot(2,2,2);
plot(f, unwrap(angle(H_lp))*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev LPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);

% High-Pass Magnitude Response
subplot(2,2,3);
plot(f, mag_hp, 'b', 'LineWidth', 1.5);
grid on;
title(['Chebyshev HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
ylim([-60 5]);
xlim([0 Fs/2]);

% High-Pass Phase Response
subplot(2,2,4);
plot(f, unwrap(angle(H_hp))*180/pi, 'r', 'LineWidth', 1.5);
grid on;
title('Chebyshev HPF Phase Response');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
xlim([0 Fs/2]);

sgtitle('Chebyshev Type-I IIR Filter Analysis');
```


Chebyshev Type-I IIR Filter Analysis
Chebyshev LPF Magnitude (Order N = 6)
Magnitude (dB)
0
-20
-40
-60
0
1000
2000
3000
4000
5000
Frequency (Hz)
Chebyshev LPF Phase Response
Phase (degrees)
0
-200
-400
-600
0
1000
2000
3000
4000
5000
Frequency (Hz)
Chebyshev HPF Magnitude (Order N = 6)
Magnitude (dB)
0
-20
-40
-60
0
1000
2000
3000
4000
5000
Frequency (Hz)
Chebyshev HPF Phase Response
Phase (degrees)
200
0
-200
-400
0
1000
2000
3000
4000
5000
Frequency (Hz)